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Investigating the Role of Different Rock Properties in Laser Perforation

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Abstract

High-power laser technology emerges as a feasible alternative to conventional perforation techniques in oil and gas wells, offering precise, clean, and debris-free perforations. The outcome of laser perforation is influenced by downhole conditions, rock properties, as well as laser properties. The objective of this paper is to evaluate how different rock properties affect laser-rock interaction and ultimately govern the perforation process.

This review provides a rigorous assessment of the impact of different rock properties and downhole conditions on the laser perforation process, leveraging an extensive range of laboratory tests, computational simulations, and observational data. The investigation focuses on properties such as in-situ stresses, rock heterogeneity, and rock hardness, all of which play a pivotal role in the laser perforation process.

Lab experiments and simulation models revealed that high-power laser drilling or perforation is a stress independent process. Changes in the confining stress magnitude or ratio did not influence the rate at which the material was penetrated. This was confirmed by the uniform temperature distribution in the tested sample among the different sensitivity cases with varying confining stress values and ratios. Heterogeneity, represented by variation in permeability and porosity, mineralogy, and presence of laminations, influenced the rate of laser perforation growth. Shale lamination had a faster thermal propagation due to the higher thermal conductivity of shale compared to the tested limestone samples. As for rock hardness, extremely hard rocks such as granite and basalt were laser perforated. Testing showed that they exhibited a faster ROP than softer and more porous rocks due to the efficiency of heat conductivity across dense and less porous rocks.

Overall, high-power lasers can provide more sustainable alternatives compared to the conventional means for different oil and gas operations such as drilling, perforation, and descaling, while tackling different technical challenges associated with the existing methods. Another distinction of high-power lasers lies in performing all of these applications with the same laser engine and system in the field with a more significantly reduced footprint than conventional practices.

Introduction

The unique characteristics of lasers make high-power laser technology a promising technology for different energy extraction applications such as drilling, perforation, scale removal, heat treatment, fracture initiation, etc. Laser perforation, a proven application in the field, has emerged as a promising alternative to conventional mechanical methods such as shaped charges, abrasive perforation, or bullet perforation. By delivering high-intensity thermal energy to the rock surface, high-power lasers can induce localized melting, vaporization, and thermal fracturing without direct mechanical contact. This technology offers the potential for cleaner perforations, reduced formation damage, and improved control over perforation geometry. The following sections provide an introductory review of high-power laser technology and laser-rock interaction.

High-Power Laser Technology

High-power laser technology has become a cornerstone in advanced material processing and subsurface applications due to its ability to deliver concentrated energy with exceptional spatial and temporal control. High-power laser systems typically operate in the kilowatt range and utilize continuous-wave (CW) or pulsed beams, depending on the application. By focusing intense optical energy onto a target surface, high-power lasers induce rapid heating, melting, or vaporization, enabling contactless material modification or removal (Zuo and Lin 2022). The development of fiber lasers and solid-state lasers has significantly enhanced the feasibility of laser deployment in industrial and downhole environments. Improvements in beam shaping, energy efficiency, and thermal management have expanded the operational range of high-power lasers (Poprawe et al. 2007). Fiber-optic delivery systems now allow laser energy to be transmitted through compact and robust conduits, making them suitable for confined or harsh settings such as boreholes.

Beyond manufacturing, the precision and controllability of high-power lasers have opened new frontiers in energy applications, including drilling, perforation, and scale removal. Their non-contact nature eliminates some of the operational challenges and offers the potential for cleaner, more sustainable operations. Additionally, laser technology experienced rapid growth, with breakthroughs in new laser designs and a consistent rise in power output, with lasers reaching the petawatt (10^{15}) order of magnitude (Flacco 2008; Hecht 2010; Duffy 2019). Continued innovation in laser sources, control systems, and diagnostics is further improving their adaptability and performance in energy and industrial applications.

Laser-Rock Interaction

High-power laser technology has emerged as a transformative tool in subsurface applications, particularly in rock perforation, drilling, and stimulation. Unlike mechanical or hydraulic methods, laser systems offer the ability to deliver concentrated thermal energy with high precision, enabling contactless rock removal and potentially enhanced operational control. The application of high-power lasers in energy applications leverages fundamental principles of thermal decomposition, melting, and vaporization to induce material removal through localized energy absorption.

When a high-power laser beam—typically operating in the kilowatt range—impinges on a rock surface, a series of coupled physical phenomena is initiated. The rock undergoes various optical transformations—some of the incident laser energy is absorbed into the material, while other portions are scattered or reflected. As the rock heats up to extreme temperatures, it can itself become a secondary radiation source, emitting blackbody radiation that further contributes to energy redistribution. Depending on the laser intensity and exposure time, this heating can result in thermal cracking, spallation, phase transitions, or direct vaporization of the material. The interaction dynamics are governed by the rock's optical properties (e.g., absorptivity, reflectivity), thermal properties (e.g., thermal conductivity, specific heat), mechanical properties (e.g., tensile strength), and downhole conditions (e.g., in-situ stresses). Additionally, significant thermal stresses may develop within the rock matrix, leading to fracture initiation. This all implies that this interaction is far from simple, as it includes a dynamic interplay of thermal, mechanical, and electromagnetic

effects occurring simultaneously. Fig 1 summarizes the multi-physics phenomena occurring during laser-rock interaction.

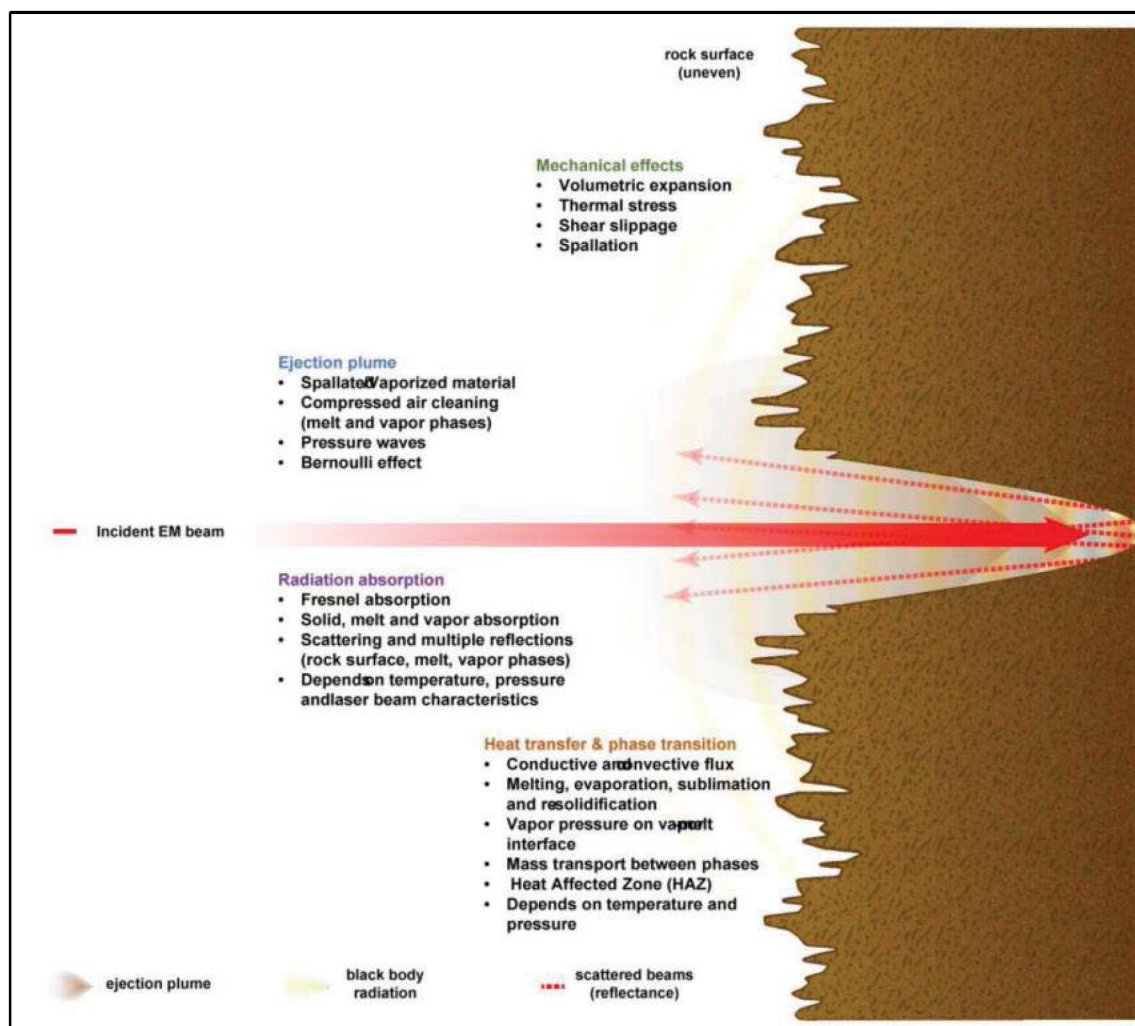


Figure 1—Reactions and changes occurring during laser-rock interaction including thermal changes, mechanical effects, phase transitions, and other phenomena (after San-Roman-Alerigi et al. 2019).

Numerous experimental and simulation efforts have been carried to improve the understanding of this reaction. Utilizing different laser types and testing a variety of rock types, lab experiments investigated the effect of lasers on rock's microgranular structure (Moavenzadeh et al. 1968), rock's absorptivity relation with rock properties (Graves and O'Brien 1998; Batarseh 2001), specific energy (SE) (Gahan et al. 2001; Ahmadi et al. 2011; Soleymani et al. 2013), and porosity and permeability enhancements by laser perforation (Batarseh et al. 2003). On the simulation side, computational models have attempted to capture the complex behavior of this interaction through various numerical and empirical approaches. Some simulation models focused on the material removal rate utilizing high-power laser (Ganesh et al. 1997), while other models studied the possibility of controlling phase transition (e.g. delaying melting) (Xu et al. 2005). Machine learning models have also been created to study key parameters influencing the laser perforation process (Keshavarzi et al. 2011). Subsequent finite element models highlighted the efficiency of laser in perforating rocks through performing parametric studies (Dini and Ahmadi 2018; Han et al. 2019; Alrashed 2022).

These studies collectively emphasize that accurate simulation of laser-rock interaction requires multidisciplinary integration—spanning optics, heat transfer, and geomechanics. This complexity makes predictive modeling both challenging and essential, particularly as the industry explores laser-based

solutions for descaling, well perforation, and drilling. Continued refinement of computational models, informed by experimental validation and field-scale observations, will be critical in advancing the reliability and deployment of laser technologies in the subsurface environment. Moreover, integration with real-time diagnostics, such as pressure and temperature sensors, thermal imaging, and emission spectroscopy, allows for adaptive control of the laser perforation process, potentially improving efficiency and safety in field deployments.

In summary, high-power lasers offer a novel, high-precision mechanism for rock interaction that can complement or replace traditional mechanical methods in select applications. Ongoing research continues to optimize the coupling of laser parameters with rock properties to enhance the feasibility and scalability of laser-based subsurface technologies. The next section explores the role of different rock properties and downhole conditions on the laser perforation process, from experimental and simulation perspectives.

Role of Rock Properties and Downhole Conditions

The effectiveness of laser perforation in subsurface formations is influenced by the intrinsic properties of the target rock and the surrounding downhole conditions. Key rock properties that govern laser-rock interaction include thermal conductivity, specific heat capacity, density, optical absorptivity, reflectivity, compressive strength (hardness). Downhole conditions include in-situ stresses, rock heterogeneity, wellbore geometry, etc. The following sections investigate the influence of in-situ stresses, rock heterogeneity, and hardness on the laser perforation process.

In-Situ Stresses

Horizontal wells are commonly drilled and positioned in a direction parallel to the minimum horizontal stress direction. This is to ensure creating a stronger and larger network of fractures with less resistance from in-situ stresses. The state of in-situ stresses (overburden stress, maximum horizontal stress, and minimum horizontal stress) usually dictates the direction and growth of fractures. This is also applicable with perforation, as the state of stress affects the perforation tunnel stability. [Batarseh et al. \(2017\)](#) conducted advanced stress testing that studied the effect of in-situ stresses on laser perforation utilizing an experimental setup that includes a biaxial frame. This setup enables the loading of up 20,000 psi confining stress on the sample being laser tested. The setup was prepared to mimic the wellbore geometry and downhole condition. Their experimental findings showed that a single perforation shot was created by laser under these extreme confining stress conditions. [Fig. 2](#) shows an image of the created perforation and the experimental setup.

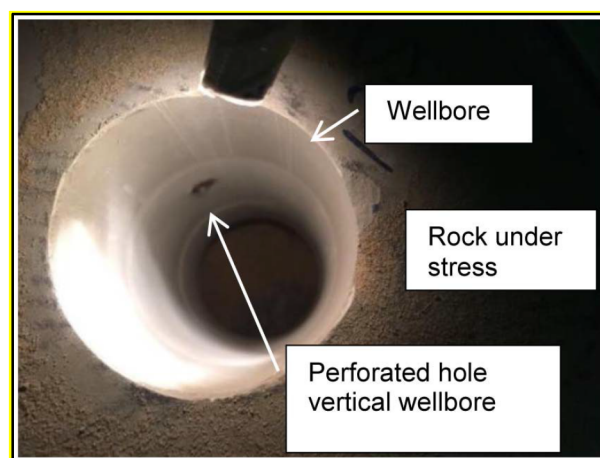


Figure 2—Experimental results of HPL perforation under confining stress conditions (after [Batarseh et al. 2017](#)).

In a numerical parametric study, [Alrashed \(2022\)](#) investigated the sensitivity of laser penetration in rocks to the presence of confining stresses. His modeling results indicated that modifying the magnitude of confining stresses or the confining stress ratio had no impact on the penetration rate. Additionally, the temperature distributions across the simulated sample of sandstone remain unchanged among the modeled sensitivity cases. [Fig. 3](#) illustrates the temperature distribution in the five simulated cases with different confining stress magnitude and ratios. This outcome demonstrates that high-power laser drilling or perforation is stress independent, meaning penetration can occur in any direction regardless of the stress state as penetration is mostly governed by the thermal propagation. This is an agreement with the experimental finding reported by [Batarseh et al. \(2017\)](#). This result is expected, as the perforation in the model is driven by temperature changes, which were not influenced by variations in confining stresses.

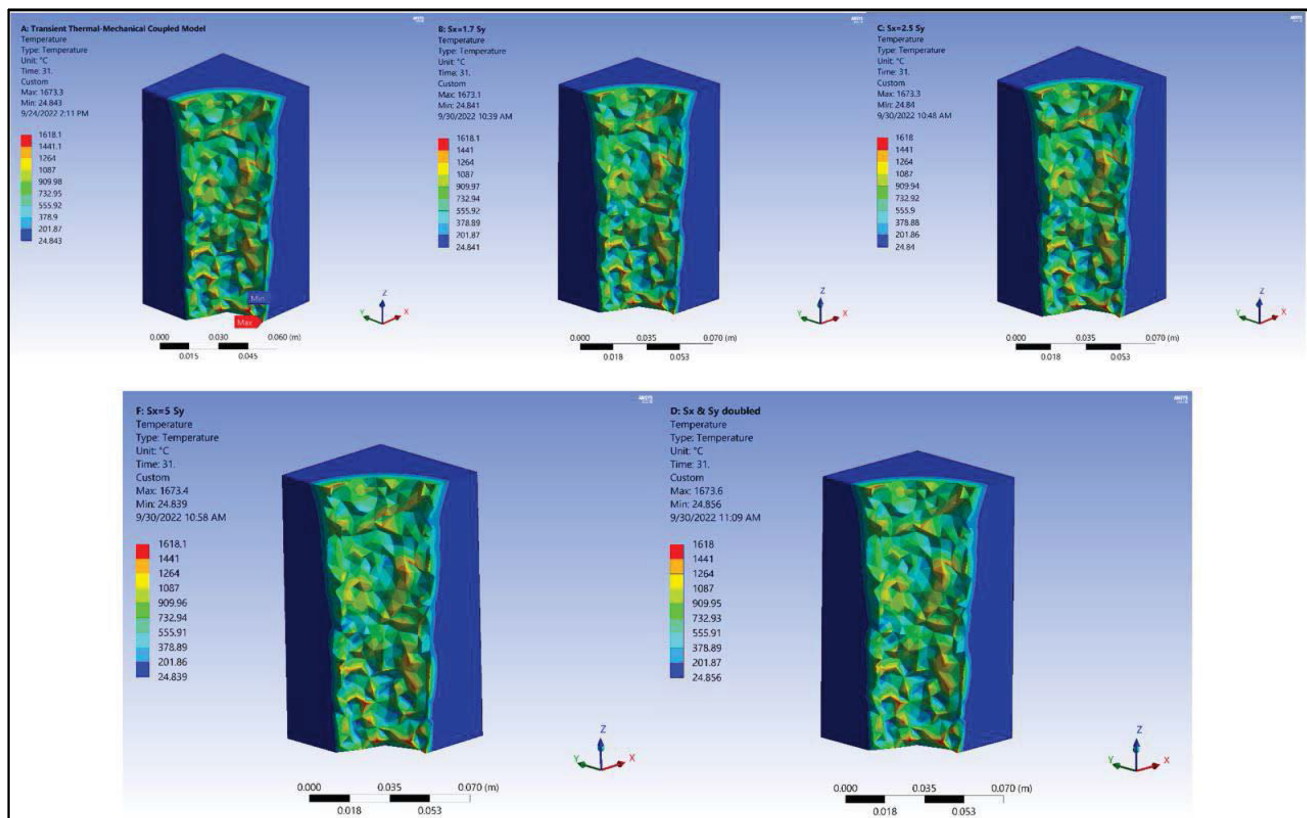


Figure 3—Temperature distribution sensitivity to change in confining stresses during laser heating (after [Alrashed 2022](#)). No notable change is observed among the five sensitivity cases.

Rock Heterogeneity

Understanding thermal energy propagation during laser perforation requires detailed characterization of the rock's microstructure and heterogeneity. Small-scale analysis of composition, grain contacts, mineral distribution, and structural features plays a role in predicting how thermal energy will travel through heterogeneous formations.

Multiple analytical methods can be integrated to build a comprehensive picture of the rock's thermal behavior. Scanning Electron Microscopy (SEM) provides insights into clay content, mineral types, grain-to-grain contact, cementation patterns, and the microstructural arrangement. Grain contacts, in particular, significantly influence thermal transfer. For instance, angular grain contacts result in reduced surface area for conduction, thereby impeding heat transfer across grain boundaries ([He et al. 2019](#)).

[Batarseh et al. \(2020B\)](#) conducted lab experiments to study rock heterogeneity and its effect on thermal propagation. They have utilized Computed Tomography (CT) scans to complement SEM by revealing

internal features such as lamination, fractures, microfractures, their orientations, and overall density distribution. These features strongly impact thermal propagation. In fractured or laminated rocks, heat transfer behavior is governed by the nature of the fracture fill material. In the experiment, a limestone sample with shale laminations was heated with a heating source placed at the edge of the rock sample in one configuration, and on the center of the sample in another configuration. Infra-red (IR) thermography was being captured during the heating process. Fig. 4 shows the IR map of temperature distribution across the heated sample. The heating pattern is not uniform across the sample. As shale has a higher thermal conductivity than limestone (3.15 W/m-k for shale compared to 1.57 W/m-k for limestone), the shale lamination reached around a temperature of 300 °C, whereas the rest of the limestone sample stayed below 280 °C, after 10 minutes of heating. Both experiments showed that in a set of multiple experiment. For example, in shale samples where fractures are filled with fine quartz, the thermal conductivity of quartz can exceed that of the surrounding matrix, resulting in localized zones of enhanced thermal transport—as illustrated in Figures 5 and 6. This leads to non-uniform thermal propagation patterns, which are a hallmark of heterogeneous formations.

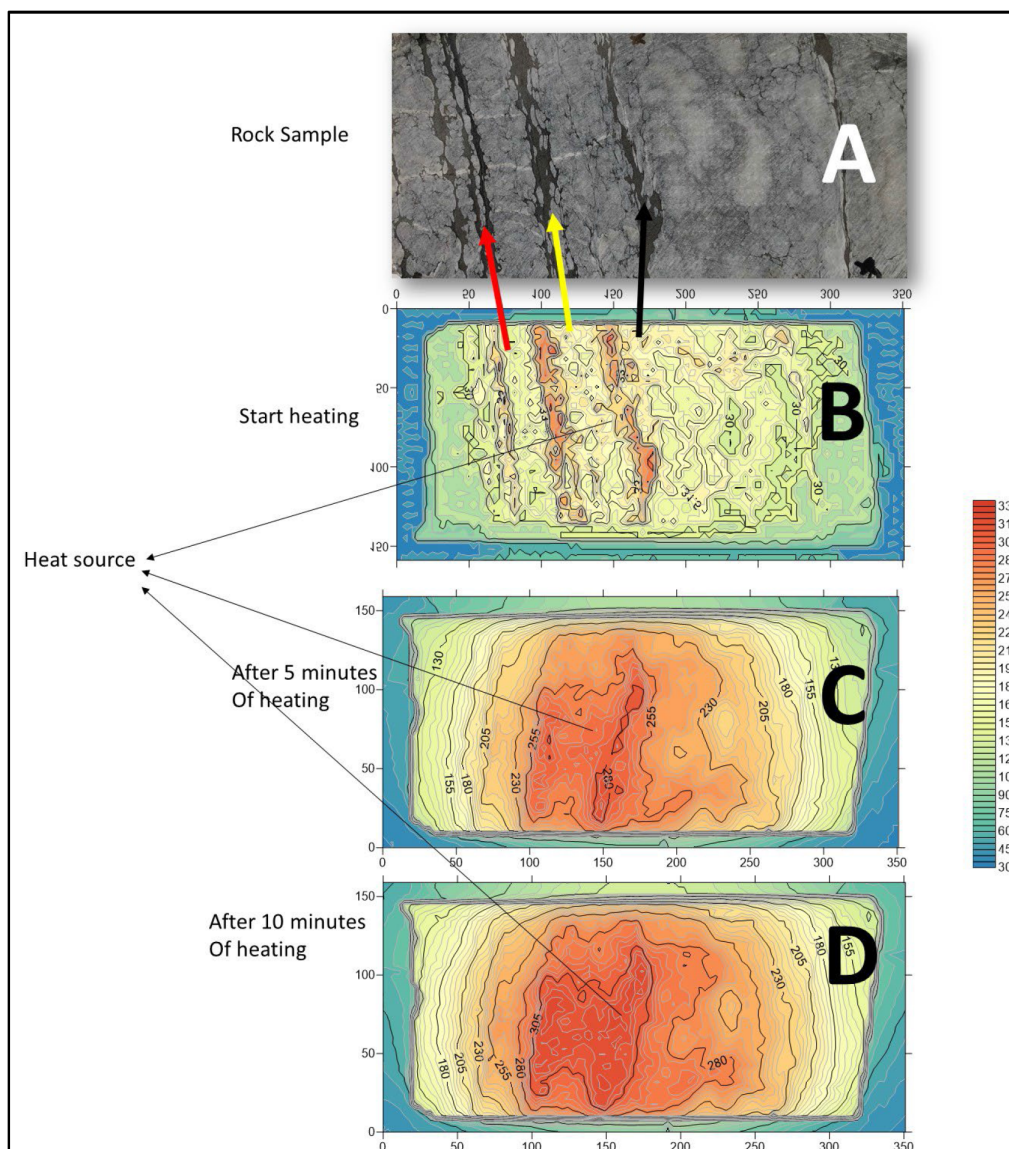


Figure 4—Temperature distribution across the heated limestone rock sample with laminated shale (after Batarseh et al. 2020B).

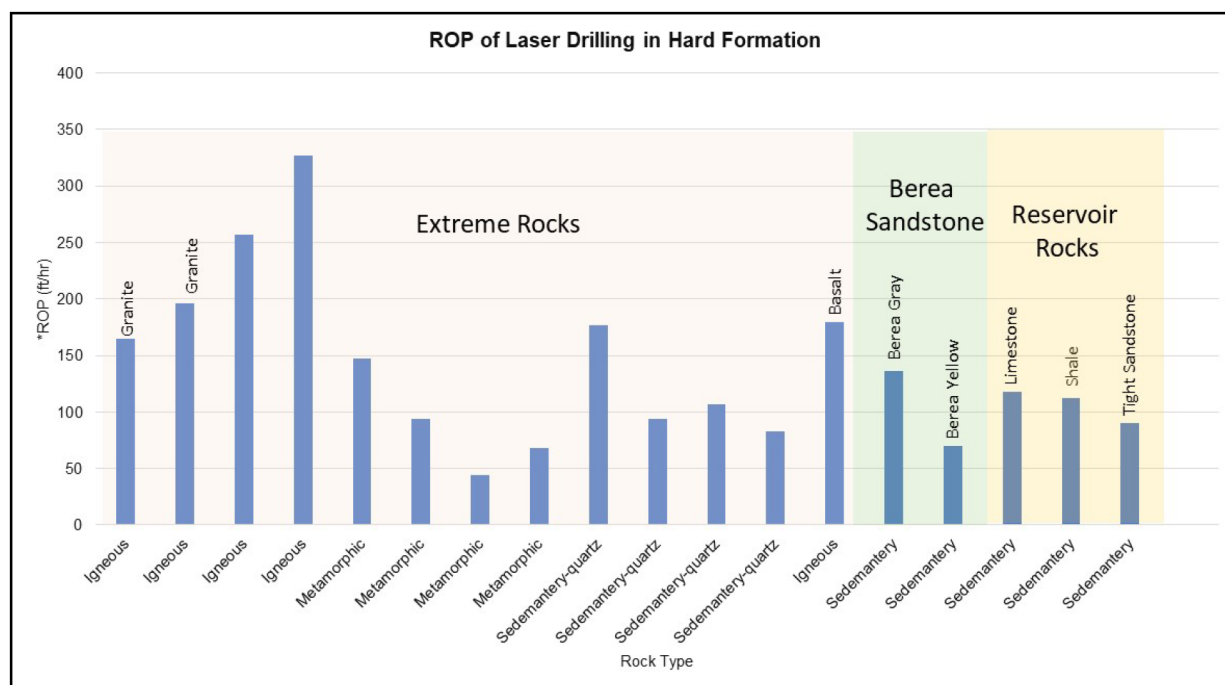


Figure 5—High-power laser rate of penetration (ROP) among different rocks including extremely hard rocks (after Batarseh et al. 2020A).

To further characterize heterogeneity, porosity and permeability variations were experimentally studied for the heated sample as well as the mineral composition. Batarseh et al. (2020B) found that minerals undergo structural transformations at elevated temperatures. Clays, for example, can collapse between 375°C and 550°C, depending on their type (illite/smectite versus kaolinite). Moreover, higher porosity and permeability affect the thermal propagation in the rock sample as the thermal conductivity is better in solid compared to solid to liquid. In laser perforation, the same principle of the effect of rock heterogeneity on the thermal propagation across the lased sample applies.

From a simulation standpoint, Han et al. (2019) investigated heterogeneity in laser perforated samples by modeling anisotropy and discontinuity. They found that propagation of perforation was asymmetrical in the heterogenous samples. Additionally, increasing discontinuities delays the propagation of perforation to the farther regions. These simulation results are in agreement with the experimental finding about the non-uniformity in thermal propagation and perforation growth in heterogenous rock samples. In summary, rock heterogeneity introduces irregularities in thermal propagation due to variations in mineralogy, microstructure, porosity, and fracture systems. Addressing these complexities is essential for accurate modeling of laser-rock interaction and for optimizing laser parameters to ensure efficient energy delivery in heterogeneous formations.

Hardness

Drilling hard rocks are typically considered challenging for conventional mechanical drilling due to their high strength and abrasiveness. Batarseh et al. (2020A) conducted a set of lab experiments on laser drilling of extremely hard formations. Pre and post laser drilling tests were performed on a variety of rock samples, including sandstone, limestone, shale, granite, and basalt, utilizing different lab equipment such as CT scans, Thermogravimetric Analyzer/Differential Thermal Analyzer (TGA/DTA), and X-ray Diffraction (XRD). Experimental results demonstrate that high-power laser drilling is effective in extremely hard formations, such as basalt and granite, regardless of their compressive strength or in-situ stress orientation. Moreover, high-power laser has shown the ability to achieve higher rates of penetration (ROP) in such hard rocks compared to softer and more porous formations. As the mechanism of laser drilling relies on thermal energy

absorbed by the rock, that gives it a fundamental advantage in drilling extremely hard rock since it is independent of the rock's compressive strength (hardness).

In dense, low-porosity rocks such as the tested basalt and granite samples, heat conduction is more efficient due to continuous solid-solid contact without any disruption in thermal conductivity by fluid or void space (no convection). This promotes deeper and more uniform thermal penetration, enabling rapid and effective ablation of the rock surface. In contrast, softer rocks such as Berea sandstone, characterized by high porosity and significant void space, exhibit slower laser penetration rates. In these materials, heat is partially dissipated through convective processes in the pore spaces, reducing the overall thermal efficiency. This is in agreement with the findings discussed in the previous section of Rock Heterogeneity, where rock samples with higher porosity and permeability exhibit discontinuous thermal propagation. Fig. 5 illustrates the laser ROP in different rocks tested in the lab, including extremely hard formations such as basalt and granite.

Summary

High-power laser technology has gone through an extensive R&D work of lab testing and numerical simulation, all prior successfully demonstrating it in the field for different applications. To expand on its field applications, continued experimentation and simulations are performed to enhance the understanding of the complex process of laser-rock interaction. This work investigates the influence of different rock parameters and downhole conditions on the laser perforation process, building up on various lab tests and numerical models. The following conclusions can be drawn from this investigation:

- Altering the magnitude of confining stress or ratio between the simulated in-situ stresses had no impact on the rate of laser penetration of the rock, confirming that laser perforation is stress independent.
- Heterogeneity, represented by changes in mineralogy or presence of laminations, create a non-uniform pattern of thermal propagation, as it is influenced by the thermal conductivity of the different components present within a heterogeneous rock sample.
- Heterogeneity, represented by variations in porosity and permeability, influences the thermal propagation induced by laser heating, as heat conduction is more efficient in less porous rocks (driven by conduction without the presence of convection heat transfer).
- The laser ROP values in extremely hard rocks such as granite and basalt were higher than those in soft rocks.

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